

WEEK-6

RISK ASSESSMENT — MCQ MASTER NOTES (PART 1)

1. Core Idea of Risk Assessment

 Always revolves around **3 questions**:

- What is the potential event?
- What can be the cause?
- What can be the consequences?

 **MCQ TRAP:** Cause $\neq$ consequence (very commonly confused)

2. Real-Life Importance (Facts to Remember)

- **2017 India road deaths: 1.47 lakh**
- Equivalent to population of **Shillong (Meghalaya)**
- **USA deaths (falls at home): 7000 annually**
 - Mostly **age > 65**

 **Key takeaway:**

 Risk is about **CHOICES**

3. Definition of Risk

 **Risk = Likelihood of harm**

Includes:

- **Acute effects (short-term)**
- **Chronic effects (long-term)**

Affects:

- **Human health**
- **Quality of life**
- **Ecological systems**

- Economic systems

📌 **MCQ KEYWORD:** Likelihood + Harm

4. Risk Assessment (Definition)

👉 Process to estimate:

- Form
- Dimension
- Characteristics of risk

Uses:

- Qualitative methods
- Quantitative methods

📌 **MCQ TRAP:** Both methods are used (not only one)

5. Risk Management

👉 **Definition:** Process by which risk is **controlled or reduced**

Steps (VERY IMPORTANT FOR MCQs)

1. Identify risk
2. Evaluate risk
3. Control risk
4. Measure risk
5. Monitor risk
6. Report risk

📌 **Order-based MCQs can come**

6. Risk in Regulatory Decisions

👉 Used as **basis for regulations**

Important Values:

- **Cancer endpoint:**
👉 1 in 1 million = 1×10^{-6}

Basis:

- **Epidemiological data**
- **Animal testing** (used when human data unavailable)

 **MCQ TRAP:** Most data comes from **animal studies**

7. Familiar Risks (IMPORTANT NUMBERS)

Reference Point:

 **1×10^{-6}**

- Below this → **Not a concern (USDA)**

Risk Levels:

- **Minimal risk:** 10^{-10} to 10^{-12}
- **Very high risk:** 10^{-2} to 10^{-1}

Examples:

- Drowning (bathtub/year): **10^{-5} to 10^{-6}**
- Airplane crash death: **10^{-5} to 10^{-6}**
- Automobile accident: **1.7×10^{-4}**
- Lightning: **10^{-3} to 10^{-4}**

 **HIGH PROBABILITY MCQs HERE**

8. Basic Principles (VERY IMPORTANT)

Risk Assessment (Basic)

- Probability that **adverse effect occurs**
- All human activities have **some risk**
- Mostly related to **chemical exposure**

Extended Definition

- Identify **chemical exposures**
- Analyze **dose-response relationship**
- Quantify risk

📌 IDEAL APPROACH = DOSE-RESPONSE BASED

🔴 9. Dose Concept (VERY IMPORTANT)

👉 Calculate:

- Amount absorbed into human body

Compare:

- Absorbed dose vs EPA safe dose

⚠ Important EPA Notes:

- For **non-carcinogens** → Safe dose exists
- For **carcinogens** → ❌ No safe dose
👉 ANY exposure = cancer risk

👉 Uses:

- Risk-specific dose for carcinogens

📌 VERY IMPORTANT MCQ POINT

🔴 10. Dose-Response Curve

👉 Relationship:

- Increasing dose → Increasing effect

Regions:

1. No-effect region
2. Threshold
3. Increasing effect
4. Maximum effect

📌 MCQ TRAP: Threshold exists for non-carcinogens

🔴 11. Risk Definitions (Part 1)

✓ Applied Dose

- Amount **available for uptake**

✓ Absorbed Dose

- Amount **actually absorbed**
- Depends on:
 - Route (ingestion, skin, inhalation)

✓ Reference Dose (RfD)

👉 Safe dose for humans

- Usually from **animal data**

📌 **KEY:** Non-carcinogens use RfD

● 12. Risk Definitions (Part 2)

✓ Risk Specific Dose (RsD)

- Dose causing **1 in 1 million risk (10^{-6})**

✓ Hazard Index (HI)

$HI = \frac{\text{Absorbed Dose}}{\text{RfD}}$
 $HI = \text{RfD Absorbed Dose}$

👉 Interpretation:

- $HI < 1 \rightarrow$ **Safe**
- $HI > 1 \rightarrow$ **Unsafe (reduce concentration)**

📌 **VERY IMPORTANT NUMERICAL MCQ**

✓ Risk Management (Advanced)

Factors:

- Cost/benefit
- Consumer needs
- Alternative chemicals

● 13. Risk Assessment Process (Drinking Water)

Step 1: Hazard Identification

- Which chemicals?
- Concentration?
- Carcinogen or not?

Step 2: Dose-Response

- Relationship between dose & effect
- Data validation
- Use RfD / RsD

Step 3: Exposure Assessment

Example:

- Concentration = 0.5 mg/L
- Water intake = 2 L/day
- Body weight = 70 kg

👉 Dose:

$$=0.0143 \text{ mg/kg/day} = 0.0143 \text{ \, mg/kg/day} = 0.0143 \text{ mg/kg/day}$$

Step 4: Risk Estimation

- Compare dose with RfD

Example:

$$HI = 0.0143 / 0.2 = 0.0715 \quad HI = \frac{0.0143}{0.2} = 0.0715 \quad HI = 0.2 \cdot 0.0143 = 0.0715$$

👉 Since $HI < 1 \rightarrow \text{SAFE}$

14. Assessing Risk (Concept)

- Use:
 - RfD (non-carcinogen)
 - SF (slope factor for carcinogen)

👉 Used to:

- Estimate risk
- Set **protective limits**

15. Cancer Risk Graph Insight

- Linear extrapolation used
- At low concentration:
 - Chemical may be:
 - **More potent OR less potent**

📌 MCQ: Linear assumption in carcinogens

16. IMPORTANT FORMULAS (HIGH WEIGHTAGE)

✓ Carcinogen (Water Standard)

$$C = \frac{TR \times BW}{CSF_o \times WC} \quad C = CSF_o \times WCTR \times BW$$

✓ Non-Carcinogen

$$C = \frac{TR \times BW}{RfD_o \times IR} \quad C = RfD_o \times IRTR \times BW$$

✓ Soil Standard (Carcinogen)

$$C = \frac{TR \times BW}{CSF_o \times IR} \quad C = CSF_o \times IRTR \times BW$$

Where:

- **TR** = Target risk
- **BW** = Body weight
- **CSF_o** = Cancer slope factor
- **WC / IR** = Intake rate

🚨 FINAL QUICK REVISION (READ BEFORE EXAM)

- Risk = **Likelihood of harm**
- Cancer limit = **10⁻⁶**
- **HI < 1 → Safe**
- **No safe dose for carcinogens**
- **RfD → non-carcinogen**
- **Dose-response → core concept**
- **Both qualitative + quantitative used**

RISK ASSESSMENT + EIA — MCQ MASTER NOTES (PART 2)

1. Risk Analysis Process (VERY IMPORTANT)

 4 Main Steps:

1. Hazard Identification
2. Dose–Response Evaluation
3. Exposure Assessment
4. Risk Characterization

 MCQ TRAP:

 Order-based question (remember sequence!)

2. Risk Assessment (Definition – Extended)

 Provides:

- Qualitative OR Quantitative estimation

Of:

- Likelihood of adverse effects

Due to:

- Exposure to specific health hazards
- OR absence of beneficial influences

 KEYWORDS: likelihood + adverse effects + exposure

3. Risk Assessment Matrix (IMPORTANT)

 Based on:

- Severity (Catastrophic → Negligible)
- Probability (Frequent → Eliminated)

Risk Levels:

- High

- Serious
- Medium
- Low

 MCQ LOGIC:

- High severity + high probability → HIGH risk
- Low probability + low severity → LOW risk

4. Risk Management (Concept)

 Comes AFTER risk characterization

Goals:

- Control toxic chemical exposure
- Leads to:
 - Law
 - Policy

 MCQ TRAP:

Risk management ≠ assessment (comes after)

5. Real-World Case Studies (VERY IMPORTANT)

 Hudson River (USA)

- PCB-contaminated sediment removal
- Started: May 2009
- Part of Superfund remediation
- Phase 2: 2012

 Viktor Yushchenko Case

- Former Ukrainian President
- Poisoned with dioxin
- Year: 2004 election campaign

 **MCQ: Chemical exposure → visible health impact**

 Libby, Montana (USA)

- Population: 3000
- Contamination: Asbestos fibers
- Source: Vermiculite mine
- One of the deadliest Superfund sites

✓ Asbestos Removal

- Requires:
 - Full-body protective gear
 - Controlled removal

📌 MCQ: Occupational safety + exposure control

🔴 6. What is Environment?

👉 Dictionary meaning: Surroundings

👉 Includes:

- All external conditions affecting:
 - Humans
 - Animals
 - Plants

✓ Components:

- Nature
- Climate, soil, water, air
- Landforms, vegetation, minerals
- Social, cultural, economic factors

📌 KEYWORD: External conditions

🔴 7. National Environmental Policy Act (USA)

👉 Environment includes:

- Physical
- Social
- Cultural
- Economic
- Aesthetic dimensions

👉 **Definition:**

- **Complete complex of factors affecting:**
 - **Life**
 - **Relationships**
 - **Survival**

📌 **MCQ: Multidimensional concept**

🔴 **8. Elements of Environment**

1. Physical Environment

- **Physical + chemical + biological factors**

2. Social Environment

- **Social + cultural life**

3. Economic Environment

- **Employment**
- **Income**
- **Economic base**

4. Aesthetic Environment

- **Historical**
- **Archaeological**
- **Landscape**

📌 **MCQ: Classification-based question**

🔴 **9. Physical Environment (DETAILS)**

Includes:

- **Land & climate**
- **Vegetation & wildlife**
- **Land use**
- **Infrastructure**
- **Air pollution**

- Noise pollution
- Water pollution

📌 MCQ: Pollution types included

10. Impact Assessment

👉 Definition:

- Systematic analysis of:
 - Problem
 - Objective
 - Alternatives
 - Likely impacts

👉 Done in:

- Structured way (EU Guidelines)

11. Why Impact Assessment?

👉 Needed because:

- Decision-makers face multiple choices
- Helps:
 - Compare alternatives
 - Ensure coherence
 - Improve policy-making

📌 IMPORTANT:

- Supports policy
- ❌ NOT a substitute for decision-making

12. Law, Policies & Institutional Arrangements

👉 Aim:

- Support environmental sustainability

Key Points:

- **Systematic procedures for:**
 - Implementation
 - Quality control
 - Compliance
- **Integrated consideration of:**
 - Biophysical
 - Social
 - Health
 - Risk impacts

Advanced Points:

- **Strategic Environmental Assessment (SEA)**
- **Sustainability integration**
- **Link with:**
 - Planning
 - Regulatory systems

 **MCQ: SEA vs EIA**

13. What is EIA? (VERY IMPORTANT)

 **Environmental Impact Assessment (EIA):**

- **Assessment of environmental consequences**
- **Before decision to proceed with:**
 - Plan
 - Policy
 - Program
 - Project

 **Includes:**

- **Positive + Negative impacts**

14. Evolution of EIA (IMPORTANT TIMELINE)

Pre-1970s:

- **Focus: Economic + technical**

1970s:

- Began in developed countries
- Environmental awareness ↑

1980s:

- Included:
 - Social
 - Institutional aspects

Late 1990s:

- Integrated approaches

🔴 15. EIA in India

👉 Developed:

- 1970s

👉 Law introduced:

- By Ministry of Environment & Forests
- Date: 27 January 1994

👉 Purpose:

- Environmental clearance for projects

📌 VERY IMPORTANT DATE

🔴 16. Why EIA? (IAIA Definition)

👉 Process includes:

- Identifying
- Predicting
- Evaluating
- Mitigating

Effects on:

- Physical
- Social

- Other systems

👉 Done before major decisions

🔴 17. EIA vs SEA

- EIA → Projects
 - (dam, airport, factory)
- SEA → Policies / Plans / Programs

📌 COMMON MCQ

🔴 18. Learning Objectives (Exam Trick Section)

After unit:

- Understand EIA as management tool
- Trace evolution
- Forecast environmental changes
- Explain SEA
- Follow environmental clearance procedures
- Conduct EIA study

📌 Sometimes direct MCQs come from this!

🔴 19. Benefits of Environmental Assessment

👉 Improves:

- Project design

👉 Past weakness:

- Poor techniques
- Ignoring findings during implementation

📌 MCQ: EA is useful but implementation matters



FINAL QUICK REVISION (PART 2)

- 4 steps: Hazard → Dose → Exposure → Characterization
- Risk management → after assessment
- Environment = external conditions
- EIA = pre-decision impact analysis
- SEA = policy level
- India EIA law = 1994
- Impact assessment ≠ decision (only supports)



FINAL ULTRA-REVISION (PART 3)

- Screening → identifies major projects
- Scoping → defines key issues + ToR
- EIS → final report
- SEA → policy level
- EIA → project level
- India clearance law → 1994
- Category A = central, B = state
- No-action alternative = mandatory



FINAL QUICK REVISION (PART 2)

- 4 steps: Hazard → Dose → Exposure → Characterization
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EIA + SEA + CLEARANCE — MCQ MASTER NOTES **(PART 3)**

1. Benefits of Environmental Assessment

Core Benefits:

- Cost-saving (design modifications early)
- Reduced project delays
- Avoided treatment/cleanup costs
- Improved project performance
- Increased project acceptance
- Avoid legal impacts & violations

 **MCQ TIP: Benefits = economic + legal + social**

2. Benefits to Local Communities

- Improved human health
- Healthier local environment
- Maintenance of biodiversity
- Reduced resource use
- Fewer conflicts over resources
- Increased knowledge, skills, pride

 **Direct MCQs possible**

3. Principle of EIA

 **General principle applies to:**

- EIA + other assessment processes

Related Processes:

- Social Impact Assessment
- Risk Assessment
- Life Cycle Analysis
- Energy Analysis
- Health Impact Assessment

- Regulatory Impact Assessment
- Species Impact Assessment
- Technology Assessment
- Economic Assessment
- Cumulative Impact Assessment
- Strategic Environmental Assessment (SEA)
- Integrated Impact Assessment

 MCQ TRAP: Many types—don't confuse names

4. Purposes of EIA

- Facilitate decision-making
- Aid development planning
- Instrument for sustainable development

 VERY IMPORTANT

5. Screening (EIA Stage)

 Purpose:

- Identify projects with significant environmental impacts

 Depends on:

- Country regulations

 MCQ: First step in EIA clearance

6. Scoping (VERY IMPORTANT)

 Identifies:

- Key significant issues

 Eliminates:

- Less important issues

 Ends with:

- Terms of Reference (ToR)

👉 Ensures:

- Focus on important impacts
- Saves time & cost

✓ Purpose of Scoping:

- Identify important issues
- Define time & space boundaries
- Gather decision-making info
- Select significant effects for detailed study

📌 HIGH PROBABILITY MCQ

● 7. Alternatives in EIA

👉 Must consider:

- Different locations
- Scales
- Processes
- Layouts
- Operating conditions
- No-action option (VERY IMPORTANT)

📌 MCQ TRAP: No-action option is always included

● 8. EIA Process (Detailed Flow)

Pre-EIA:

1. Project screening
2. Scoping

Main EIA:

3. Baseline data
4. Impact prediction
5. Impact evaluation
6. Mitigation measures
7. Summary/report

Final:

- 8. Review & decision-making**
- 9. Approval / rejection**

Post-EIA:

- 10. Monitoring & management**

 **Sequence-based MCQs**

9. Hierarchy in EIA Studies

- **Site selection studies**
- **Rapid / comprehensive studies**
- **Regional studies**
- **Carrying capacity studies**

10. EIA Cycle (IMPORTANT)

- **Pre-feasibility**
- **Feasibility**
- **Design & engineering**
- **Implementation**
- **Monitoring & evaluation**

 **EIA runs through entire project lifecycle**

11. Environmental Impact Statement (EIS)

 **Document containing:**

- **All EIA findings**

 **Helps:**

- **Decision-makers**
- **Regulators**

 **Can lead to:**

- **Project modification OR rejection**

✓ EIS Structure:

Part 1:

- **Methods + key issues**

Part 2:

- **Preliminary studies**
- **Site description**
- **Proposed activities**

Part 3:

- **Detailed impacts:**
 - **Land use**
 - **Water**
 - **Air**
 - **Ecology**
 - **Noise**
 - **Socio-economic**

📌 MCQ: 3-part structure

🔴 12. Impact Indicators

👉 Based on:

- **Statutory standards**

Example:

- **Air Quality Index (AQI)**

Categories:

- **Good**
- **Better**
- **Best**
- **Acceptable**
- **Unacceptable**

📌 MCQ: Indicators = legal standards

● 13. Strategic Environmental Assessment (SEA)

👉 Definition:

- Formal, systematic process for:
 - Policies
 - Plans
 - Programs (PPP)

👉 Includes:

- Written report
- Public decision-making

✓ Tier Concept:

- Policy → Plan → Program → Project

📌 MCQ: Hierarchical relation

● 14. Rationale & Scope of SEA

1. Strengthening EIA

- Complements project-level EIA

2. Cumulative Effects

- Better than EIA for large-scale impacts

3. Scope:

- Covers:
 - Regional planning
 - Development programs

● 15. EIA vs SEA (VERY IMPORTANT TABLE)

EIA	SEA
Reactive	Proactive
Project-based	Policy/Plan-based
Specific project	Regions/sectors
Direct impacts	Cumulative impacts
Mitigation focus	Sustainability focus
Narrow scope	Broad scope

 **VERY COMMON MCQ**

16. Environmental Clearance Procedure (India)

Steps:

- 1. Screening**
- 2. Scoping**
- 3. Baseline data + impact prediction**
- 4. Alternatives + mitigation + EIS**
- 5. Public hearing**
- 6. Decision making**
- 7. Monitoring**

 Sequence question possible

17. Documents Required for Clearance (VERY IMPORTANT)

Key Documents:

- Application form
- Project report
- EIA report
- Risk assessment report
- Site clearance
- Consent from SPCB
- NOC from authorities
- Utility availability proof
- Hazardous material details
- Wildlife clearance (if needed)
- Rehabilitation plan (if displacement)
- Forest clearance
- Coastal regulation clearance
- Public hearing details
- Filled questionnaire

 **Stakeholders:**

- Local residents
- NGOs
- Environmental groups

 **HIGH CHANCE MCQ**

18. Environmental Clearance Categories

Category A:

- Central Government (MoEF)
- Expert Appraisal Committee (EAC)

Category B:

- State Government (SEIAA)

- State Expert Committee (SEAC)

 **IMPORTANT:**

 If no SEIAA → treated as Category A

FINAL ULTRA-REVISION (PART 3)

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FINAL EXAM STRATEGY (IMPORTANT)

 Focus MOST on:

- Definitions (Risk, EIA, SEA)
- Numbers (10^{-6} , $HI < 1$)
- Sequences (steps, process)
- Comparisons (EIA vs SEA)
- Keywords (RfD, RsD, HI, CSF)

DESIGN, SUSTAINABILITY & LCA — MCQ MASTER NOTES (PART 4)

1. Design and Sustainability (Core Idea)

 What we:

- Design
- Create
- Consume

 Directly affects:

- Environmental impact (SUBSTANTIALY)

 Common Terms:

- Sustainable Engineering
- Sustainable Design
- Design for Environment (DfE)

 Move beyond:

- Carbon footprint
- Water footprint

 **MCQ:** DfE = design considering environment

2. Environmental Considerations in Design

 Must be:

- Key part of design
- Always addressed

 Linked with:

- Life Cycle Analysis (LCA)

 **MCQ:** LCA is central to sustainable design

● 3. Can Engineers Change Environmental Footprint?

👉 YES:

- Need tools to identify **better design options**

👉 Otherwise:

- No purpose of engineering improvements

📌 **Conceptual MCQ possible**

● 4. Two Levers in Engineer's Hand (VERY IMPORTANT)

1. Material & Resource Inputs

- Keep **minimum**

2. Design Decisions

- **Design for disassembly**
- **Design for maintainability**
- **Design for longevity**

📌 **HIGH PROBABILITY MCQ**

● 5. Over-Packaging Problem

👉 Example:

- Excess packaging → unnecessary waste

📌 **MCQ:** Design inefficiency increases environmental impact

● 6. Why Should We Care?

Environmental Issues:

- Rising sea levels
- Ozone hole
- Greenhouse gases
- Ocean plastic

- Soil fertility
- Water supply

 **Direct factual MCQs**

7. Example: Fish Contamination

 Warning:

- Fish contain **contaminants (e.g., mercury)**

 Risk:

- Cancer + health issues

 **MCQ:** Bioaccumulation concept

8. Resource Use Per Day

 **> 50 kg/person/day** resource use

Inputs:

- Metals
- Minerals
- Forestry products
- Agriculture
- Energy materials
- Construction

Outputs:

- Wastes
- Air emissions
- Products in use

 **IMPORTANT NUMBER**

9. Sustainability Definition (VERY IMPORTANT)

 **Sustainable development:**

- Meets present needs
- WITHOUT compromising future generations

 **Classic MCQ (definition-based)**

10. Life Cycle Thinking

 Consider entire life of product:

Stages:

1. Concept
2. Material extraction
3. Manufacturing & packaging
4. Transportation
5. Retail/use
6. Disposal (landfill/incineration)
7. Recycling

 **KEYWORD:** Cradle → Grave

11. Electric vs Diesel Example

 Electric:

- Looks clean locally
- ✗ But pollution may shift to power plants

 **MCQ:** Hidden impacts (system thinking)

12. Environmental Folklore

 Decisions based on:

- Beliefs
- Experience
- Media

✗ Not always:

- Scientific

👉 Leads to:

- Impact transfer (not reduction)

📌 **VERY IMPORTANT CONCEPT**

🔴 **13. Life Cycle Assessment (LCA)**

✓ **Definition:**

👉 Scientific method to assess:

- Environmental impacts of product/system/service

👉 Standards:

- ISO 14040
- ISO 14043

✓ **Expanded Definition:**

👉 Compilation & evaluation of:

- Inputs
- Outputs
- Environmental impacts
 - 👉 Across entire life cycle

📌 **HIGH PROBABILITY MCQ**

🔴 **14. LCA Study Steps (VERY IMPORTANT)**

1. Goal & Scope definition
2. Inventory analysis (LCI)
3. Impact assessment (LCIA)
4. Interpretation

📌 **Sequence-based MCQ guaranteed**

🔴 **15. LCA Phases Nature**

👉 Iterative process

- Steps can be repeated/refined

📌 **MCQ TRAP:** Not linear

● 16. Goal & Scope Definition

👉 Questions:

- Purpose of LCA?
- Target audience?

● 17. Inventory Analysis (LCI)

👉 Questions:

1. Function & functional unit?
2. Boundaries?
3. Required data?
4. Assumptions?
5. Limitations?

● 18. Impact Assessment (LCIA)

👉 Measures:

- Environmental
- Social
- Economic effects

● 19. Interpretation

👉 Final step:

- Reduce impacts
- Draw conclusions
- Recommend actions

● 20. Example: Pencil Comparison

Wooden Pencil:

- Wood, rubber, graphite

Mechanical Pencil:

- Plastic (PE, PP)
- Oil-based materials

👉 Includes:

- Manufacturing
- Transport
- Use
- Disposal

📌 **MCQ:** Functional unit comparison

21. Carbon Footprint Example

👉 Total lifecycle emissions:

- Include:
 - Raw materials
 - Manufacturing
 - Transport
 - Usage
 - Disposal

📌 **MCQ:** Not just production stage



FINAL ULTRA-REVISION (PART 4)

- Design → affects environment directly
- 2 levers → **materials + design decisions**
- LCA → **cradle to grave analysis**
- LCA steps → **Goal → Inventory → Impact → Interpretation**
- Sustainability → **future generations concept**
- 50 kg/day resource use
- Environmental folklore → **not scientific**

FINAL EXAM BOOST (VERY IMPORTANT)

 **MOST ASKED FROM THIS UNIT:**

- **LCA definition**
- **LCA steps (order)**
- **Sustainable development definition**
- **Two levers in design**
- **Life cycle stages**
- **Iterative nature of LCA**

LCA ADVANCED + FUNCTIONAL UNIT — MCQ MASTER NOTES (PART 5)

1. Function vs Functional Unit (VERY IMPORTANT)

✓ **Function:**

 **Service provided by a system**

- What it actually does

✓ **Functional Unit:**

 **Quantified version of function**

- Gives numerical value
- Used for comparison

✓ **Example:**

- Function = Writing
- Functional Unit = 1 meter of writing

 **MCQ TRAP:**

Function $\neq$ Functional Unit

2. Items to Consider in LCA System

Inputs:

- Energy
- Materials
- Labor

Outputs:

- Products (electricity, materials)
- Waste
- Emissions
- Co-products

📌 MCQ: System = inputs + outputs

3. LCA Thinking Example (Paper Plate vs China Plate)

👉 Questions to ask:

- What is the function?
- Functional unit?
- Resources used?
- Production impact?
- Waste generation?
- Washing impact (for reusable)?
- Data needed?

📌 Conceptual MCQ (very common)

4. Environmental Impact Categories

✓ Global Warming Potential (GWP)

- Greenhouse gases trap heat
- Examples:
 - CO₂, CH₄, NO₂, ozone

✓ Abiotic Depletion

- Use of non-living resources

✓ Human Toxicity Potential

- Harm to humans from chemicals

✓ Land Use

- Land required

5. More Impact Categories

✓ Eutrophication

- Nutrient increase (N, P)
- Leads to:

- Algae growth
- Oxygen depletion
- Death of aquatic life

✓ Acidification

- Caused by:
 - Acid rain
 - Fuel pollution
- Effects:
 - Low pH
 - Soil/water damage

✓ Others:

- Smog
- Energy use
- Solid waste
- Oil use
- Water use
- Mercury

📌 MCQ: Match category with effect

6. Comparison Example (Light Bulbs)

👉 Compare:

- Compact Fluorescent vs Incandescent

📌 Requires:

- Same functional unit (lighting output)

7. Life Cycle of Product (IMPORTANT)

👉 Includes ALL processes:

- Resource acquisition
- Manufacturing
- Distribution
- Use

- End-of-life

 **KEY:** Anywhere + anytime processes

8. LCA = Accounting Exercise

 **Tracks:**

- Environmental interventions
- Changes in natural systems

 **Covers:**

- Entire life cycle

9. Life Cycle Inventory (LCI)

 **Inputs:**

- Iron ore
- Crude oil
- Water
- Solar energy
- Land use

 **Outputs:**

- CO₂
- SO₂
- PM
- VOC
- NO_x
- Metals
- Pesticides

 **Other:**

- Radiation
- Heat
- Noise

 **MCQ:** LCI = data collection stage

🔴 10. Functional Unit & Reference Flow (VERY IMPORTANT ⚠️)

👉 Product comparison is based on:

- Functional unit

👉 Why needed?

- Products may have different performance

✓ Example:

- Milk carton vs returnable bottle
- Compare:
 - 1000 liters delivery

📌 KEY IDEA:

👉 Reference flow = amount needed to fulfill functional unit

🔴 11. Light Bulb Example (VERY IMPORTANT)

Function:

👉 Lighting

Functional Unit:

👉 Example:

- Providing 700 lumens for 10,000 hours

Reference Flow:

- Incandescent:
 - 10 bulbs + 600 kWh electricity
- Fluorescent:
 - 1 bulb + 14 kWh electricity

Key Parameters:

- Lifetime
- Watt/lumen ratio

 **HIGH PROBABILITY MCQ**

12. Inventory for Light Bulb

Steps:

1. Product system
2. Functional unit
3. Reference flows
4. Unit processes
5. Elementary flows

 **Includes:**

- Raw materials
- Energy
- Emissions

13. Life Cycle Inventory Complexity

 **Facts:**

- Hundreds of flows
- Hard to interpret

 **Brain limit:**

- Can handle ~7 parameters

 **Simplification:**

- Ignore processes < 0.1% impact

 **MCQ TRICK**

14. Databases in LCA

 **Example:**

- Ecoinvent database

Contributors:

- EMPA
- ETH
- PSI
- etc.

 MCQ: LCA uses databases

15. LCA in Organizations

 Used by:

- Designers
- Product managers
- LCA experts
- EMS specialists

 Applications:

- Ecodesign
- Strategy
- Benchmarking
- Product declaration

16. Impact Assessment Structure

Flow:

 Inventory → Midpoints → Endpoints

Midpoints:

- Global warming
- Ozone depletion
- Acidification
- etc.

Endpoints:

- Human health
- Ecosystem quality
- Resources

 **VERY IMPORTANT MCQ**

17. Life Cycle Impacts

 **Inputs → Outputs → Impact categories → Single score**

Categories:

- Global warming
- Ozone depletion
- Acidification
- Eutrophication
- Human toxicity
- Ecotoxicity

18. IMPACT2002+ Method

 **Combines:**

- Midpoints + Endpoints

Damage Categories:

- Human health
- Ecosystem quality
- Climate change
- Resources

 **Advanced MCQ**

19. Final Interpretation (VERY IMPORTANT)

 **Final stage:**

- Combine results
- Draw conclusions
- Suggest improvements



FINAL ULTRA-REVISION (PART 5)

- Function ≠ Functional Unit
- Functional unit = comparison basis
- LCA = cradle to grave + accounting system
- LCI = data collection stage
- Midpoint → Endpoint
- Ignore <0.1% impact
- 700 lumens example = IMPORTANT



FINAL MASTER REVISION (ALL UNITS)



MOST REPEATED MCQ AREAS:

- Risk formulas (HI, RfD, RsD)
- EIA steps + definitions
- SEA vs EIA
- LCA steps + functional unit
- Impact categories
- Numerical values (10^{-6} , HI<1, etc.)

LCA DEVELOPMENT + APPLICATION — MCQ MASTER NOTES (PART 6)

1. Development of LCA (IMPORTANT TIMELINE)

 Evolution highlights:

- **1960s**
 - REPA study (Coca-Cola, USA)
- **1984**
 - Environmental report on packaging (Switzerland)
- **1991**
 - 1st workshops by **SETAC**
- **1997**
 - **ISO 14040 standards introduced**
- **2000–2012**
 - UNEP + Life Cycle Initiative
- **2010**
 - Social LCA guidelines

 **Important Organizations:**

- **SETAC** (Society of Environmental Toxicology and Chemistry)
- **ISO**
- **UNEP**
- **Ecoinvent Center**
- **European Platform on LCA**

 **MCQ:** ISO 14040 = LCA standard

2. LCA in Decision Making (Waste Management)

 Used to:

- Compare **alternatives**
- Select **best option**

 **Case Examples:**

- Peel Region (Canada)
- Auckland (New Zealand)

- Compost facility odor control

 **MCQ:** LCA supports decision-making

3. Waste Management Systems (IMPORTANT FLOW UNDERSTANDING)

System 2 (APEFW + Landfill)

- MSW → Facility
- Outputs:
 - Metals → market
 - Energy → market
 - Fly ash + bottom ash → landfill

System 3 (Mass Burn Facility)

- Waste burned → energy recovery
- Outputs:
 - Electricity/steam
 - Metals recovered
 - Residuals → landfill

System 4 (RDF Combustion)

- RDF → boiler
- Outputs:
 - Energy
 - Recyclables
 - Residues → landfill

 **MCQ TIP:**

 All systems end with **some landfill waste**

4. Organic Waste Case Study (VERY IMPORTANT)

 Location:

- **Auckland, New Zealand**

Key Findings:

- **Anaerobic digestion (food waste)**
 - 👉 Better than composting
- For mixed waste:
 - 👉 **Dry anaerobic digestion + composting**
 - 👉 Better than composting alone

📌 HIGH PROBABILITY MCQ

🔴 5. General Conclusions (FINAL TAKEAWAY)

👉 Topics covered:

- Risk assessment
- Environmental risk assessment
- Life Cycle Analysis (LCA)

🔥 FINAL ULTRA-REVISION (ALL CONTENT)

🔥 MOST IMPORTANT (REVISION IN 2 MIN):

- Risk = **Likelihood of harm**
- $HI < 1 \rightarrow \text{Safe}$
- Cancer risk = 10^{-6}
- EIA = **Project level**
- SEA = **Policy level**
- India EIA law = **1994**
- LCA = **Cradle to grave**
- Steps = **Goal $\rightarrow$ Inventory $\rightarrow$ Impact $\rightarrow$ Interpretation**
- Functional unit = **comparison basis**
- Design:
 - Minimize resources
 - Design for longevity
- Case study:
 - **Anaerobic digestion > composting**

